INTELLIGENT SYSTEMS / SEMINAR 1

Reinforcement learning with the Astronaut

Learning to fly, avoid asteroids and collect oxygen with a Q-table

Andrea Kitanovska (89262058) | Hristijan Kochovski (89262125)

FLIGHT
100.0%
Games completed without a collision

+ ASTEROIDS
98.6%
Games completed without a collision

+ OXYGEN
96.8% / 5.32 tanks
Games completed / average tanks per game

Five independently trained agents. Asteroid and oxygen results use 100 new test games per agent.

01 How the agent learns

We first teach the astronaut to stay in the air. We then add asteroids and, finally, oxygen tanks. Each harder stage builds on the Q-values learned earlier.

A SMALL TABLE OF POSSIBLE SITUATIONS

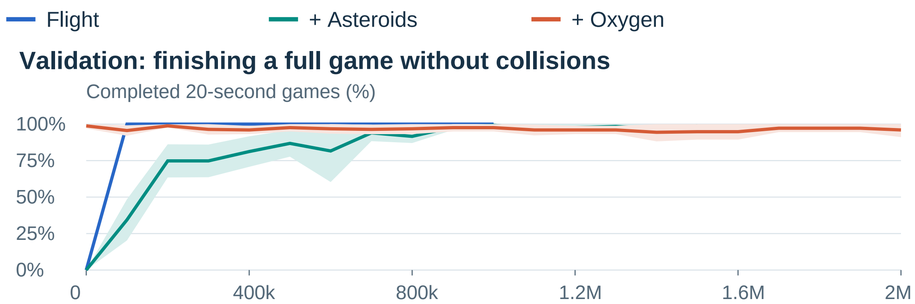
Flight Height and vertical speed	**72 states**
+ Asteroids Flight + asteroid arrival time and offset	**2,880 states**
+ Oxygen Asteroids + tank position and availability	**103,680 states**

For each situation, the table stores a value for coasting and a value for thrusting. The agent explores both actions and updates their values after each attempt.

```
Q(s,a) <- Q(s,a) + alpha * error
```

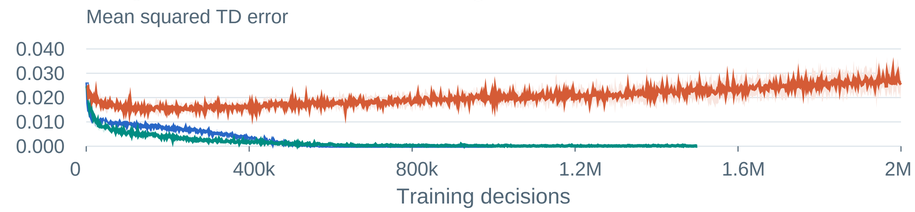
The error compares the current value with the reward and the best next value. Actions last up to six frames. A tank earns +1; a collision earns -1; other frames earn 0.

02 What changes during training



At each checkpoint, we test without exploration: one flight game, or 50 games with objects. Higher lines mean fewer collisions. Later stages start with earlier skills.

How the Q-value prediction error changes



Flight and avoidance errors fall. Oxygen error stays above zero even when collection improves; this does not show loss convergence.

03 How it plays after training

These are separate test games, used after selecting each policy. Values show the mean +/- standard deviation across five agents.

Flight
Completed: 100.0 +/- 0.0%
Average game: 20.00 seconds

+ Asteroids
Completed: 98.6 +/- 2.6%
Average game: 19.88 seconds

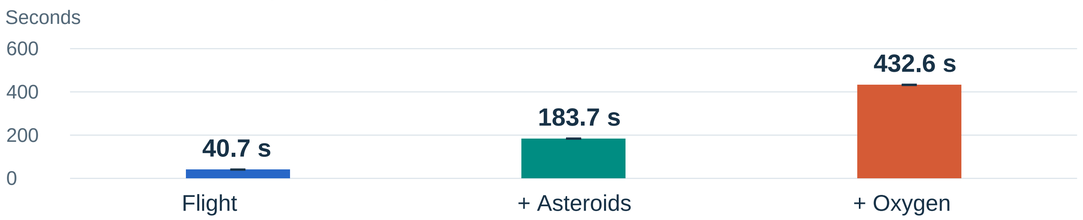
+ Oxygen
Completed: 96.8 +/- 3.1%
Tanks/game: 5.32 +/- 0.61

The oxygen agents collected 2,660 tanks across 500 test games: 5.32 tanks per game on average.

93.2% of games finished with at least three tanks. The shaded bands in the learning graphs show variation between agents.

TRAINING TIME: HOW LONG DID LEARNING TAKE?

Time spent learning this stage



Total time, including earlier stages

